DATA PREPROCESSING REPORT

Preparation of the HAM10000 dermoscopy corpus

What was done to the data, in what order, and what changed

DM-AHR-ViT · Explainable Vision Transformer for Skin Lesion Classification Using Dermoscopy Images

Research prototype only. Not clinically validated and not intended for diagnosis, triage or patient care.

Stage	Images	What it means
Collection as received	11,720	everything downloadable from ISIC collection 212
After cohort + integrity cleaning	10,015	the confirmatory HAM10000 cohort used for all results
Quarantined	1,705	extra images with no usable label; never enter a split
Lesion groups after deduplication	7,470	the unit that gets split and counted

In one sentence

The corpus starts as 11,720 JPEG images of identical geometry and is cleaned down to 10,015 labelled images forming 7,470 lesion groups. Cleaning removes 1,705 unlabelled extras, merges 21 confirmed near-duplicate pairs, and produces four matched input variants plus 1,022 synthetic evaluation pairs.

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1. The raw data, before anything is done

The starting point is the client's ISIC Archive collection 212. Every image the collection exposes was downloaded and then measured, rather than assumed: file size, pixel dimensions, colour mode, encoding and a SHA-256 checksum were recorded per file. The table below describes the corpus exactly as received.

Property	Value	How it was established
Images downloaded	11,720	collection listing, cursor-paged to exhaustion
Pixel dimensions	600×450	identical for all 11,720 images
Colour mode	RGB	verified per file
Encoding	JPEG	verified per file
Decode status	11,720 of 11,720 ok	every file opened successfully
Median file size	26,599 bytes	confirmatory cohort
Images with a diagnosis label	10,015	present in HAM10000 metadata
Images without a label	1,705	not in HAM10000 metadata

Two facts here shape everything that follows. First, the collection is larger than the 10,015 images the research proposal cites, because collection 212 also exposes images outside the labelled HAM10000 set. Second, the extras carry no diagnosis, so they cannot be trained on or scored — they are not merely redundant, they are unusable for a supervised task.

> **Important distinction**
> 1,705 extra images are not corrupt or low quality. They are valid dermoscopic photographs that simply have no label in this corpus. They are quarantined rather than deleted, so the decision is reversible and auditable.

2. The preprocessing pipeline, step by step

Preprocessing runs as an ordered chain. Each step reads files, writes new artifacts and records what it did, so any later count can be recomputed from disk. Nothing is performed silently and no image is dropped without a logged reason.

#	Step	Operation	Output
1	Integrity audit	Open every file; verify decodability, dimensions, bit depth; hash each image	manifest + checksums
2	Cohort definition	Match against HAM10000 metadata; label the remainder as extras	10,015 confirmatory
3	Exact duplicate removal	SHA-256 collision sweep	0 duplicates found
4	Near-duplicate detection	dHash candidates, then pixel confirmation on a common grid	21 pairs confirmed
5	Lesion grouping	Merge lesion identifiers with confirmed duplicate links	7,470 groups
6	Stratified split	70/15/15 per class, groups never straddling partitions; hash frozen	10,015 images assigned
7	Hair screening	Multi-orientation black-hat/top-hat detector, threshold frozen	8,993 hair-present
8	Arm construction	Classical removal, 128 px control round-trip, synthetic pairs	four arms + pairs

Why deduplication was done in two stages

A perceptual hash (dHash) proposes candidate duplicates cheaply, but on dermoscopy it is far too permissive: similar skin backgrounds collide, and it proposed 70,062 candidate pairs among 10,015 images. Treating those as duplicates would have merged unrelated patients. Each candidate was therefore re-checked pixel by pixel on a common 96×96 greyscale grid and kept only if mean absolute difference stayed at or below 4/255 and correlation at or above 0.985. 21 pairs survived. All 21 of them already shared a lesion identifier, so this step confirmed that no unseen cross-lesion duplicates existed rather than changing the grouping: the 7,470 lesion groups equal the unique lesion identifiers in the metadata. The check is still worth having — it is the evidence that grouping by lesion does not miss a duplicate pair the metadata does not link — but it should not be credited as additional leakage control.

Why the split is by lesion, not by image

HAM10000 contains several photographs of the same lesion in different sessions. If those photographs were scattered across training and test, the model would be asked to

recognise a lesion it had already seen, and the test score would measure memorisation rather than generalisation. Splitting at the lesion level makes an honest test set, and the duplicate check above establishes that no unlinked duplicates slipped through. The split is hashed and frozen before any model is trained, so it cannot be quietly adjusted afterwards.

Why the hair threshold is frozen

The hair detector has tunable thresholds. They were calibrated on training and validation sources only, then frozen before a single test image was processed. Had the threshold been adjusted after seeing how the test set behaved, the test set would have influenced a design choice, and every later result would be contaminated by it.

3. Before and after, in numbers

The comparison below is between the corpus as received and the corpus after cleaning and arm construction. Every figure is recomputed from the manifests, not carried over from the proposal.

Measure	Before	After	Change
Images	11,720	10,015 in cohort	−1,705 quarantined
Usable labels	10,015	10,015	unchanged
Duplicate pairs merged	70,062 candidates	21 confirmed	pixel-verified
Splitting unit	10,015 images	7,470 lesion groups	groups, not images
Train / val / test	—	7,011 / 1,502 / 1,502	70 / 15 / 15
Hair status known	no	8,993 present, 1,022 absent	measured
Classical-removal images	—	10,015	Arm B
Diffusion-removal images	—	10,015	Arm C
Resolution-control images	—	10,015	Arm A*
Synthetic evaluation pairs	—	1,022	with true hairless reference

Class distribution after splitting

Class	HAM10000 expected	Cohort	Train	Val	Test	Share
akiec	327	327	229	49	49	3.3%
bcc	514	514	360	77	77	5.1%
bkl	1,099	1,099	769	165	165	11.0%
df	115	115	81	17	17	1.1%
mel	1,113	1,113	779	167	167	11.1%
nv	6,705	6,705	4,693	1,006	1,006	66.9%
vasc	142	142	100	21	21	1.4%
total	10,015	10,015	7,011	1,502	1,502	100%

Observed counts matched the published HAM10000 counts exactly (yes), which independently confirms that the cohort definition recovered the intended dataset rather than a similar one. Every class holds 14.8–15.0% of its images in each split, so the rare classes (`df`, `vasc`) retain a small but non-zero test presence.

Hair burden in the cleaned cohort

Hair coverage statistic	Value
Images screened	10,015
Screened hair-present	8,993 (89.8%)
Screened hair-absent	1,022 (10.2%)
Mean covered fraction	0.109
Median covered fraction	0.061
90th percentile	0.277
Maximum	0.989

Hair is not a marginal nuisance in this corpus: at this threshold 89.8% of images screen as hair-present, and the worst images are almost entirely covered. That prevalence is why armour against hair artefacts is worth studying — and also why the detector's precision matters so much, discussed in section 5.

Effect of cleaning on image fidelity

On the 186 held-out synthetic pairs — images with a genuine hairless reference — fidelity can be measured rather than judged. The unprocessed hairy input is scored as a floor: a method that scores below the image it was given has made that image worse.

Condition	PSNR (dB)	SSIM	Interpretation
Hairy input (floor)	22.13	0.520	untouched image, hair in place
Arm B — classical	35.73	0.918	much closer to the hairless reference
Arm C — diffusion	16.31	0.130	below the untouched input in this run

Arm B improves fidelity substantially, consistent with classical inpainting doing what it is designed to do on hair-sized structures. Arm C scores below the unprocessed input in this run, which is a real result about the model that was tested, not a scoring error. The cause is a defect in the training script, not the inference path: the released DM-AHR weights reach PSNR 26.8 / SSIM 0.684 through the same adapter, while the checkpoint trained here reaches 15.0 / 0.079, because the script gave the network an inverted noise-level signal. Its output is noisy and structurally unrelated to the target rather than blurred. Arm C in this study is a failed replication — see section 7.

4. Visual before-and-after

Numbers do not show what happened to an individual lesion, so the pipeline also writes contact sheets. The three figures below are excerpts from those sheets. They were inspected by the engineer who built the pipeline; they have not been reviewed by a dermatologist, and are described here in exactly those terms.

4.1 Hair detection

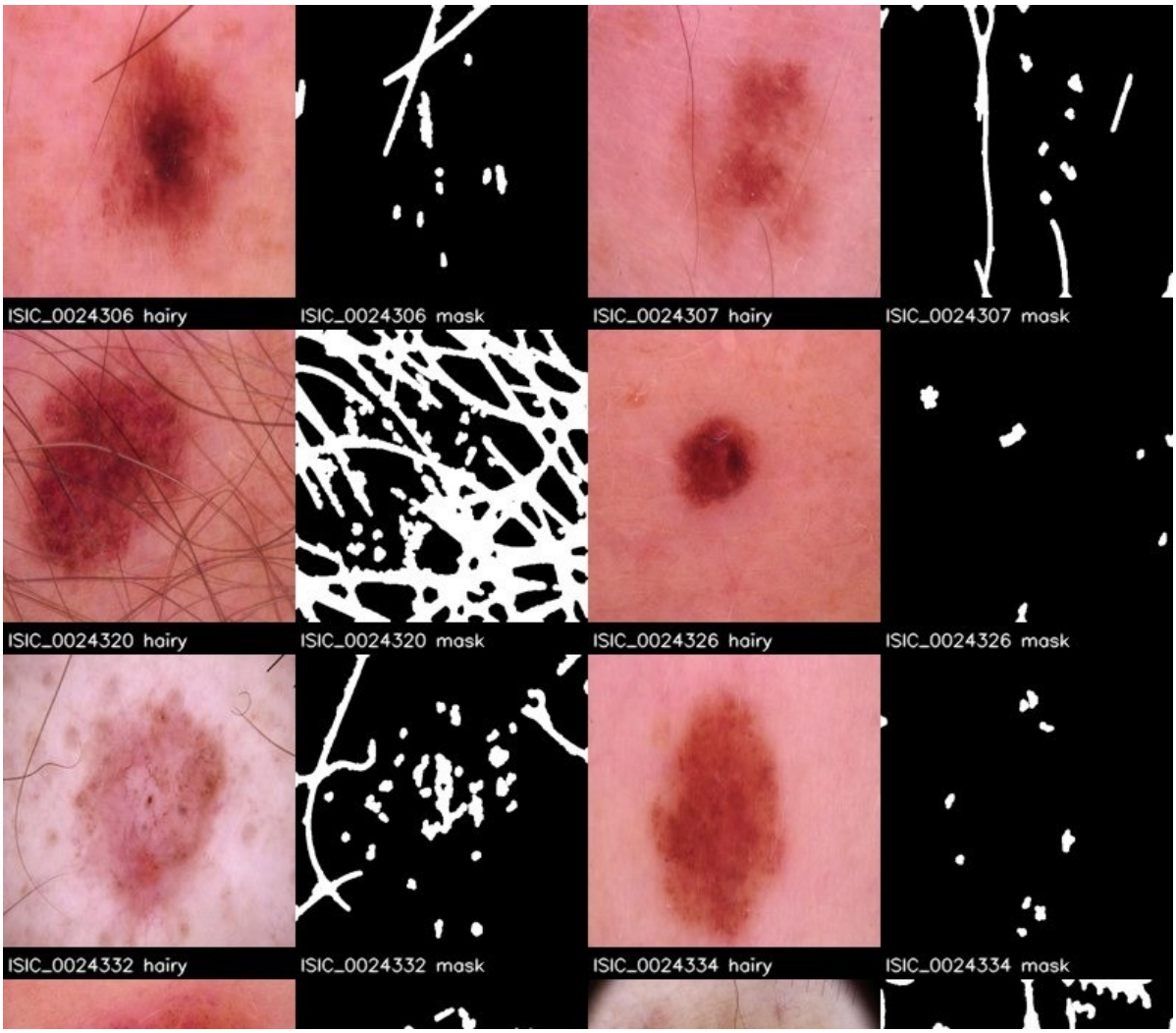


Figure 1 — Hair detection. Each pair shows the raw image and the binary hair mask the detector produced for it.

What is visible: on sparse cases the detector cleanly isolates individual hairs as thin strands while leaving the lesion untouched. On the heavily-haired case it correctly marks a dense field of overlapping hairs. Two limitations are visible in the same figure. First, where hair overlays the lesion, the mask also covers part of the lesion itself, because dark lesion pigment and dark hair are similar in the filter response. Second, on a few clean images the mask contains scattered small specks that are texture or noise rather than hair. Both effects are the expected failure modes of a morphological detector, and both are why the detector's output is used to decide whether to act, with inpainting restricted to the marked pixels.

4.2 Classical hair removal (Arm B)

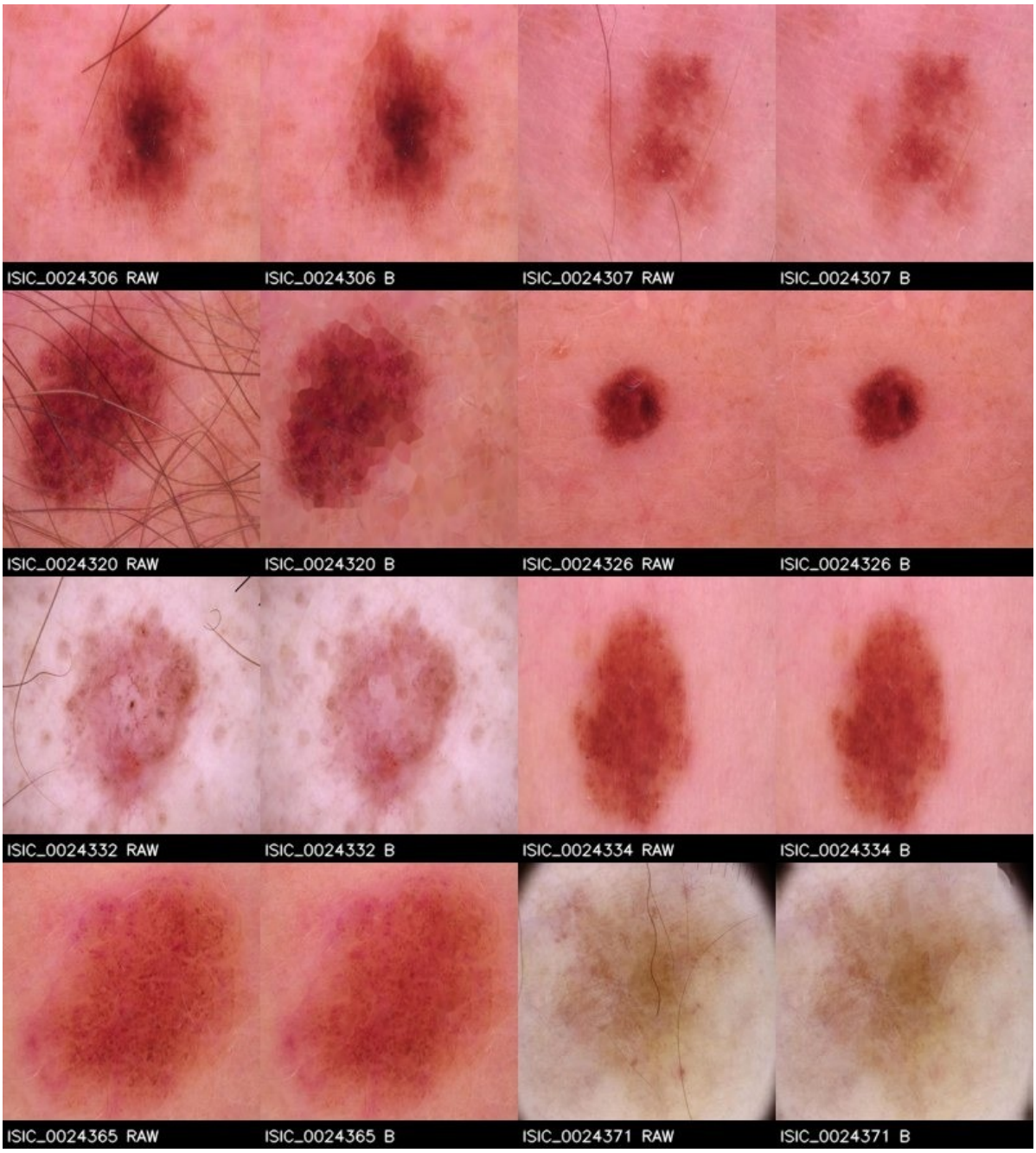


Figure 2 — Arm B, DullRazor-style classical removal. Raw image beside the processed result for the same lesion, in four pairs.

What is visible: fine hairs are removed with the lesion boundary preserved, which is the intended behaviour and consistent with Arm B's high PSNR. On the heavily-haired case the result is different in character — the hairs are gone, but the region they crossed is left slightly blotchy, because interpolation fills the removed pixels with neighbouring colour rather than reconstructing structure. This is the classical method's defining limitation, and it is the reason a generative approach is worth testing at all.

4.3 Diffusion hair removal (Arm C)

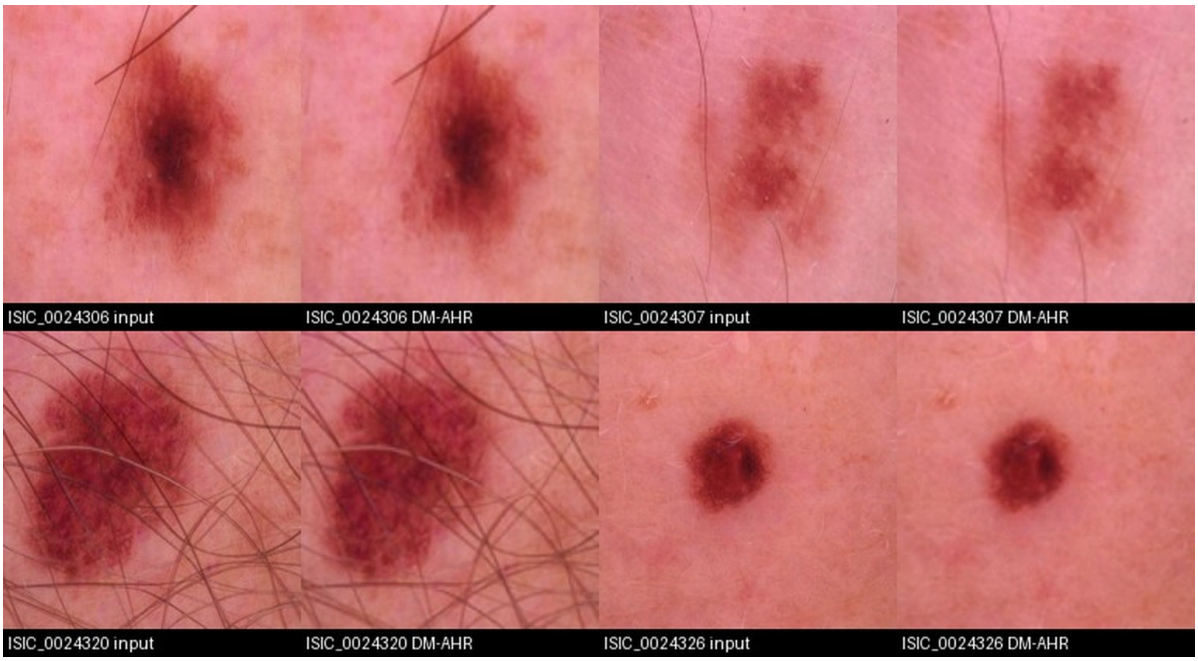


Figure 3 — Arm C, DM-AHR diffusion removal. Input beside the model's output for four lesions.

What is visible: hair is largely suppressed, but the result is degraded in a way that is not simple blurring — the lesion loses definition and the surface takes on a mottled, low-contrast appearance, and on the heaviest case faint hair remnants persist. This matches the measured PSNR and SSIM, and the diagnosis confirms the cause: the output carries far more high-frequency energy than the reference while its global contrast falls, which is what a model that has not learned the mapping produces. It is a finding about the specific checkpoint trained for this study, which was trained far below the reference budget — not evidence that diffusion hair removal cannot work.

5. What cleaning changed and what it cost

5.1 Effects that strengthen the study

- **Honest cohort —** removing 1,705 unlabelled images keeps every reported metric inside the labelled cohort, so no result depends on images with no ground truth.
- **Leakage control —** splitting by lesion keeps every photograph of one lesion inside a single partition, so the test set cannot reward memorising a lesion the model already saw.
- **Duplicate verification —** the 21 pixel-confirmed duplicate pairs all already shared a lesion identifier, so this verifies the metadata's grouping rather than extending it.
- **Frozen parameters —** the frozen split hash and the frozen hair threshold mean the test set could not have influenced any design choice.
- **Matched arms —** all four arms share identical manifests, transforms and seeds; only the hair-removal condition differs.
- **Measurable fidelity —** 1,022 synthetic pairs give a genuine hairless reference, which is what makes PSNR and SSIM meaningful instead of decorative.

5.2 Costs and trade-offs, stated plainly

- **High screening prevalence —** at the frozen threshold, 89.8% of images are treated as hair-present. That is a high rate and is reported as a limitation rather than hidden. It means most images are processed, which maximises cost and risk of altering images that needed no alteration.

- **Screening is not validated —** the proposal specifies the manually annotated hair masks of Lama et al. (ChimeraNet) as an independent reference for screening quality, and those masks were not obtained. This is a skipped validation step, not a gap in the data. The detector was calibrated on synthetic pairs and frozen, which protects the test set but does not validate it against real hair.
- **False positives propagate —** the heavy-hair case in Figure 1 shows the mask covering lesion pigment too. Classical inpainting then fills those pixels, which can smooth lesion structure along with the hair.
- **Resolution change —** Arm C operates at 128 px natively, so its inputs are upsampled to the 224 px the classifier expects. Arm A* isolates this by round-tripping originals through the same resize, and that control was inconclusive — meaning the resolution change alone does not explain Arm C's deficit.
- **Storage —** processed images are written to disk at their working resolution, which triples storage requirements relative to the raw corpus and is why processed arms live on a volume rather than in the repository.

6. What happens at training time, not in advance

Not all preprocessing is applied to files. Two steps happen in memory, per image, every epoch. It matters for the write-up because these are part of the method even though they appear nowhere on disk.

Stage	Training split	Validation and test splits
Geometric	RandomResizedCrop to 224 px (scale 0.75–1.0, ratio 0.9–1.11), horizontal flip, vertical flip, rotation ±15°	Resize to 256 px, centre crop to 224 px
Photometric	Colour jitter: brightness 0.15, contrast 0.15, saturation 0.10, hue 0.02	none
Value scale	to tensor in [0, 1], then ImageNet mean/std normalisation	identical
Class imbalance	inverse-frequency class weights from the training split only, in the loss	not applied (no weighting at evaluation)

Augmentation is deliberately applied only to training images. Validating on augmented images would make the validation signal noisy and the early-stopping decision arbitrary. The identical recipe is used for all four arms, so the comparison is not confounded by augmentation differences.

> **A deviation from the proposal, disclosed**
> The proposal specifies rotation up to ±90° with explicit zoom and shift. The implemented recipe uses ±15° rotation with a random resized crop that provides scale variation instead of a separate shift parameter. The difference is recorded in the project decision log. Because the recipe is identical across all four arms, the internal comparison remains valid; what it cannot claim is that the exact stated augmentation was used.

7. Limitations

- **Hair screening is unvalidated —** the Lama et al. manually annotated hair masks specified by the proposal were not obtained, so the detector's accuracy on real hair is unverified. Behaviour on synthetic pairs is not evidence about real hair.
- **Classical removal destroys texture —** the fix is pixel-level interpolation, so lesion texture under removed hair is not recovered, only approximated.
- **Arm C is a failed replication —** the training script gave the network an inverted noise-level signal (t/T instead of the sampler's sqrt(alphas_cumprod_prev)), and the checkpoint was also trained well below the reference iteration budget and sampled at half the reference steps. The released weights work through the same inference path, so the failure is the training run, not the adapter. The fixed script did not reach usable sampling at the budgets tried. Arm C is a failed replication, not evidence about the published method.
- **Synthetic pairs are a proxy —** synthetic hair is a proxy for real hair. It gives a true reference image, which real hair cannot, but its appearance differs from genuine terminal hair.
- **Lesion masks are single-rater —** the HAM10000 segmentations were created by a single dermatologist (Tschandl P), network-initialised and then verified, corrected or replaced by hand in FIJI (Tschandl et al., Nature Medicine 2020). They are dermatologist-produced but single-rater, with no inter-rater agreement available, so they are one tracing rather than consensus ground truth.
- **Licensing is unresolved —** HAM10000 is CC BY-NC 4.0, and the DM-AHR repository's README and LICENSE disagree (CC BY-NC 4.0 against Apache-2.0). Commercial use is unresolved and needs a rights decision.
- **Single corpus —** the images are a single public corpus from one collection. Nothing here establishes behaviour on another scanner, population or acquisition protocol.

8. How to reproduce

Each step is a standalone command. They can be run on Modal, on a local GPU, or in a Colab GPU runtime; the data steps need no GPU at all.

Step	Command	Needs
Download	python scripts/00_download_data.py	CPU, ~7 min
Audit + split	python scripts/01_audit_and_splits.py	CPU, ~1.5 min
Screening + arms + pairs	python scripts/02_hair_pipeline.py	CPU, ~10 min
Everything after	python scripts/03… 10_finalize.py	GPU

Verified independently: the three data steps were re-run from a completely fresh download with no Modal present, and reproduced the frozen split hash fda56cf3c555e6d2…, the same 7,470 lesion groups, the same 8,993 hair-present count and the same 1,022 synthetic pairs. That byte-level agreement is the strongest available evidence that the preprocessing is deterministic and portable.

Every number in this document traces to a file: data/manifests/data_manifest.csv, splits.csv, hair_screening.csv, synthetic_pairs.csv and near_duplicates_confirmed.csv, plus artifacts/logs/paired_psnr_ssim_test.csv. The figures come from artifacts/panels/. This document is generated by scripts/13_preprocessing_report.py, so it can be rebuilt from the artifacts rather than edited by hand.